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Response to the Replication Project by David Reinhard Regarding Study 1 of

Förster, J., Liberman, N., & Kuschel, S. (2008). The effect of global versus local processing styles on assimilation versus contrast in social judgment. *Journal of Personality and Social Psychology, 94(4)*, 579-599.

In this study, David Reinhard aimed at replicating Study 1 of our 2008 JPSP paper.

Specifically, in this replication several effects were expected to replicate:

First, one would have expected a replication of a classic priming effect (Srull & Wyer 1979, see Higgins, Rholes & Jones, 1977) that has been demonstrated many times (for reviews see Higgins, 1996). More specifically, an effect of aggression priming on person evaluation was expected, so that an experimental group primed with aggression would evaluate an ambiguously aggressive target person, John, to be more aggressive than the control group that was not primed with aggression.
Priming should only affect the evaluative dimension of aggression but not the other, control dimensions, which were unrelated to aggression. Although the original studies were conducted in the US, successful replications in other countries were also conducted.

We used the original material translated from the US version and were first unable to replicate the effect in Bremen/Germany in several pretests. We then changed the paradigm slightly, because we realized that our participants did not perceive the original person description as aggressive – this could have been due to cultural differences in what is perceived to be aggressive behavior.
One pre-condition for getting priming effects is *applicability*. According to this basic principle, the content of the prime has to fit the content of the target. In other words, an aggressive prime should not have an influence on person evaluation if the target person does not have any features of aggression (Higgins, Rhole & Jones, 1977). To illustrate, if participants are primed with aggression and are then asked to evaluate a woman, priming effects sometimes fail to obtain, to the extent that aggression does not apply to participants' view of women (Banaji, Hardin & Rothman, 1993). In this classic counter-example of priming, a misfit between target and priming was established through the stereotype of females that typically does not include aggressiveness.

Because in our Bremen sample the target was not perceived as aggressive, we needed to *create* fit within the classic aggression scenario for our specific sample. We therefore slightly enhanced the aggressiveness of the description by adding more aggressive behaviors. Fortunately, in this situation we were able to replicate the classic effect, reflecting the fact that an aggressive prime can affect the evaluation of an ambiguously aggressive target. In our experiment, aggression was measured with two ratings, and the two dimensions were correlated at .63. Thus, it was also justified to calculate a sum score of the two aggression-related ratings.

The main purpose of the study, however was to find an interaction with global/local processing. Indeed, we found the priming effect to be moderated by another variable, a so-called local/global priming procedure. We found the priming effect to be stronger in the global group compared to the local and the control groups, and we found a contrast effect in the local condition showing that participants primed with aggression perceived John to be less aggressive than participants primed with words not related to aggression. As expected, we

obtained a significant two-way interaction between aggression priming and processing mode priming.

David Reinhard aimed to replicate the study using our re-translated stronger version of the aggressive target in the US. He also needed to change the priming procedures- whereas in Germany we had a German city map, he used state maps from the US. Thus, the study was a *conceptual* replication of our findings – a straight replication was impossible due to cultural differences.

He found a main effect of processing styles in that in the global condition, participants found the target to be more aggressive. However, he failed to replicate the interaction we found. Importantly, he also failed to replicate the classic aggression priming (main) effect that was a *pre-condition* for finding an additional interaction. In general, it appears that his sample found John's behavior more aggressive than our sample, as can be seen in higher aggressiveness ratings in all conditions.
This however, could be a reason why no contrast effect was found in the local condition – by the logic of priming, extremely aggressive behavior should not be influenced by a subtle priming procedure. Notably, *ambiguity* of the target is another important pre-condition for priming effects to occur (Higgins, 1996); if the person description is clear and unambiguous, priming usually has no effects.

Moreover, in the cultural context of the present replication attempt, the two dimensions of aggression did not correlate as expected (r = .17), indicating that the sample understood the questions differently than our sample.

Last but not least the local/global priming procedure did not work as expected, as a manipulation check reflected: for 38 out of 70 participants the priming did not work – they simply used a different processing style than the manipulated one.

Thus, we agree with the author that in his study several preconditions, especially *target ambiguity* and *applicability* to replicate our effects were not met. We are optimistic that future studies will be able to replicate our findings (see some replications below). However, as we see in this example, several pretests have to be done in order to give a replication a decent chance. One should for example pretest whether the target is perceived as ambiguous, whether relevant aggression judgments are correlated, and whether the prime fits the target. It is a pre-condition to first show that the classic, well-replicated semantic priming effect works before further moderation of our global/local priming procedure can be shown. Finally, one should make sure that the global/local priming procedure works so that participants actually use the processing style that was induced.

In general, we appreciate the replication work and we think that in general, there was a chance to find the expected effects. However, materials are sometimes perceived differently by different samples and thus adjustments have to be made to meet the psychological conditions that are needed to obtain certain effects.

Conceptual Replications of our Study:
Our JPSP 2008, shows that assimilation and perception is affected by processing styles. This follows from our *global and local processing model*, or GLOMO, predicting that a global processing style enhances similarity perception to a standard compared to a local processing style. Our studies were consistent with the hypotheses showing for example that participants perceived Steffi Graf to be more similar to them if they had looked at the gestalt of a stimulus

set before. For participants who had looked at the details before, the same standard appears to be more dissimilar to them, compared to a control group – this in the end would result in assimilation effects on person perception and self perception. We believed this to be the case because global processing is more inclusive and should therefore lead to integration whereas a local processing style should be more exclusive and lead to differentiation.

In a different paper, we could show that people in general perceive objects (different Rodin sculptures), animals (dolphins and sharks) and people (from different ethnicities) to be more similar, if primed with a global compared to a local processing style (Förster, 2009).

Expanding on or notion, Van Dellen, Sanders, and Fitzsimons (2012) showed that local processing enhances differentiation of healthy and unhealthy foods when these options are presented in the same context. Etkin & Ratner (2013) argued that distance vs. proximity (which leads to zooming in or out and thus also induces global versus local processing) and compatibility between the degree of similarity among means to goal attainment increases goal-directed motivation. Consistent with their idea, six studies demonstrate that consumers are more motivated and willing to pay for means to goal attainment in the near term (related to local processing) when they plan to use a set of different (vs. similar) means. In contrast, consumers are more motivated and willing to pay for means to goal attainment in the long term (global processing) when they plan to use similar (vs. different) means. These effects are driven by the ease of processing differences (similarities) when considering the near (far) future. This effect, too, builds on our 2008 paper and extends its logic.
Other results consistent with our model were found in a variety of domains, including health and fitness, saving money, and academic performance. To illustrate, Righetti, Finkenauer & Rusbult (2011) showed that regulatory focus, a variable related to global/local processing (see Studies 8a and 8b in our 2009 JEP:G paper) increased similarity perception with a participants' partner, which in turn improved relationship satisfaction (see for a different conceptual replication, Nussinson, Seibt, Häfner, & Strack, 2011). Moreover, McCrea, Wieber, Myers (2012) showed that global processing enhances perceived similarity to ones' own stereotype (such as whether one is female or male), in that participants identified faster self-stereotypic information. Moreover, Mok and Morris (2012) found evidence for the integration process that is assumed to underlie similarity perception in people with bicultural identities: Global processing enhanced an integration of two cultural identities while local processing impaired such integration. Global processing also increased the correct deciphering of meanings that speakers' attempted to convey by ambiguous sentences to listeners; in other words global processing allowed receivers to attend more perceptively to the larger communicative context, thereby facilitating successful communication (Woltin, Corneille & Yzerbyt, 2012). Demonstrating the influence of global/local processing on inclusion in a different domain, Krüger, Fiedler, Koch & Alves (2014) found that people make use of wider response categories when they judge distant and abstract as compared with close and concrete stimuli. Finally, Mukherjee, Srinivasan & Manjaly (2014) found that a global processing style fits approach oriented message appeals and fosters monetary allocation towards charities. There are many other papers that incorporate and build on some aspects of the theory on the relation between global (vs. local) processing and similarity (vs. dissimilarity) perception.

In sum, we firmly believe in the theory that generated the predictions that were tested in the paper in question. We dare to hope that this theory will continue to inspire researchers, and that psychologists in both applied and basic sub-disciplines will continue to find it useful.

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